



DEPARTMENT OF CHEMICAL ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY SRINAGAR

Bioresource Technology

Assignment Plus Question Bank-1

S. No.	Contents				COs
1.	Discuss the objective of the course, Bioresource Technology.				CO1-CO4
2.	Write a brief note on the outcomes (COs) of Bioresource Technology.				CO1-CO4
3.	Explain 'Bioresource Technology' and elaborate recent trends in it.				CO1-CO4
4.	What are the challenges and opportunities in 'Bioresource Technology' ?				CO1-CO2
5.	Define, explain and classify 'Bioresources'.				CO1
6.	Enlist the areas of activities of 'Bioresource Technology'.				CO1-CO4
7.	Discuss the significance of bioresources in healthcare: an Indian perspective.				CO2
8.	<i>Miscanthus</i> × <i>giganteus</i> ultimate analysis on mass basis (as received) is given as:				CO4
	Components	As received	Dry basis	Dry ash free basis	
	Carbon	43.10			
	Hydrogen	5.27			
	Sulfur	0.12			
	Nitrogen	0.34			
	Oxygen	40.43			
	Ash	2.23			
	Moisture	8.51			
	Total	100			
	Calculate on dry and dry ash free basis.				
9.	Calculate chemical formula subscript for <i>Miscanthus</i> × <i>giganteus</i> with the help of following information:				CO4
	Elements	Wt.%	Atomic wt.	No. of Atoms	
	Carbon	47.31	12		
	Hydrogen	5.78	1		
	Sulfur	0.14	32		
	Nitrogen	0.37	14		
	Oxygen	44.09	16		
	Chlorin	0.04	35		
	With the help of the following equation and Table given in Q. No.8, calculate HHV on wet basis, dry basis: $\text{HHV}_{\text{wet basis}} = 0.35C + 1.18H + 0.1S - 0.02N - 0.01O - 0.02A$				
10.	Explain, what is biomass and discuss its significance as bioresource.				CO1-CO2
11.	Describe the important properties of biomass considered significant for commercial application.				CO2



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12.	The lignin contents and higher heating values (HHV) for some of the biomass samples are given in the following Table:										CO4
	Biomass Samples	Lignin (L) Measured (wt.%) ^{db}		HHV (wt.%) ^{db} (MJ/kg)		HHV (wt.%) ^{daf} (MJ/kg)		HHV (Calculated) (MJ/kg)		Difference	
	Corn Stover	14.4		17.8		18.5		17.7		-0.8	
	Corncob	15.0		17.0		17.2		17.8		+0.6	
	Sunflower Shell	17.0		18.0		18.8		18.0		-0.8	
	Beech Wood	21.9		19.2		19.5		18.4		-1.1	
	Ailanthus Wood	26.2		19.0		19.4		18.9		-0.5	
	Hazelnut Shell	42.5		20.2		20.5		20.1		-0.4	
	Wood Bark	43.8		20.5		20.8		20.1		-0.7	
	Olive Husk	48.4		20.9		21.6		21.0		-0.6	
Walnut Shell	52.3		21.6		22.2		21.4		-0.8		
13.	With the help of the above data, develop a mathematical model which correlates higher heating values and the lignin contents.										CO4
	After finding the mathematical correlation between HHV and L in question (12), determine the square of correlation coefficient (R ²) and also calculate percentage of average error, and what is the root mean square error (RMSE)?										
14.	Note: The correlation: $RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\text{Observed value} - \text{Predicted value})^2}$ may be used.										CO4
	The proximate and ultimate analyses results of some of the bioresources are given in the Table below:										
Biomass		Proximate analysis				Ultimate analysis ^{daf}				References	
	M	VM ^{db}	FC ^{db}	A ^{db}	C	H	N	S	O		
Pine chips	7.6	72.4	21.6	6	52.8	6.1	0.5	0.09	40.5	Masia (2007)	
Poplar	6.8	85.6	12.3	2.1	51.6	6.1	0.6	0.02	41.7	Miles et al. (1995)	
Sawdust	34.9	84.6	14.3	1.1	49.8	6	0.5	0.02	43.7	Tillman (2000)	
Willow	10.1	82.5	15.9	1.6	49.8	6.1	0.6	0.06	43.4	Moilanen (2006)	
15.	db: Dry basis daf: Dry, ash-free basis, M:Moisture, VM: Volatile Matter, A: Ash, FC:Fixed Carbon										CO4
	Using the correlations: HHV (MJ/kg) = (0.3536×FC + 0.1559×VM-0.0078×Ash) and HHV(kJ/kg) = (3.55×C ² -232×C-2230×H + 51.2×C×H + 131× N + 20600) based on proximate and ultimate analyses, respectively, predict the lignin contents present in the bioresources by using the correlation developed for the data in question (12).										
Estimate the values of HHVs for the bioresources given in question (12) by using the correlations of question (14) and repeat for finding the mathematical model. Note: referred data in the literature may be used for proximate and ultimate analyses.											